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Big Data Analytics in Cloud Computing

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CSA1558: CLOUD COMPUTING AND BIG DATA ANALYTICS

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Technical Presentation



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Introduction

What is Big Data?

- Extremely large, fast-growing, and varied datasets that traditional systems cannot handle.
- Characterized by the 5 V's: Volume, Velocity, Variety, Veracity, Value.

What is Cloud Computing?

Delivers on-demand storage, computing power, and services over the internet with scalability and pay-as-you-go pricing.

Big Data Analytics in the Cloud

Using cloud platforms to store, process, and analyze massive datasets to extract actionable business insights.



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Core Technical Explanation

Why Cloud for Big Data?

- Distributed storage and parallel processing handle massive datasets.
- Elastic scaling adds or removes resources based on workload.
- Managed services reduce infrastructure and maintenance overhead.
- Supports frameworks such as Hadoop, Apache Spark, and cloud-native data warehouses.

Working Process

Data Sources → Ingestion → Cloud Storage → Distributed Processing → Analytics & ML → Insights



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Big Data–Cloud Architecture

Components of the Architecture

1. Data Sources

Transactions

Social media

Sensors and logs

2. Ingestion Layer

Apache Kafka

Apache Flume

IoT gateways

3. Cloud Storage & Processing

Data lakes (HDFS / S3)

Apache Spark, Hadoop

Managed clusters (EMR)

4. Analytics & Visualization

Machine learning models

BI dashboards

Reporting tools

Architecture Diagram



Data Sources → Ingestion → Cloud Platform → Application Layer → End Users



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Benefits of Cloud-Based Big Data Analytics

Scalability

Easily scales storage and compute for growing datasets.

Cost Efficiency

Pay-as-you-go pricing reduces upfront infrastructure cost.

Real-Time Insights

Enables fast, near real-time data processing and alerts.

Elasticity

Resources expand or shrink automatically with demand.

Advanced Analytics

Supports AI, machine learning, and predictive analysis.



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Real-World Applications

Retail & E-Commerce

Personalized recommendations
Demand forecasting

Healthcare

Patient data analytics
Disease outbreak prediction

Banking & Finance

Fraud detection
Risk analysis

Manufacturing

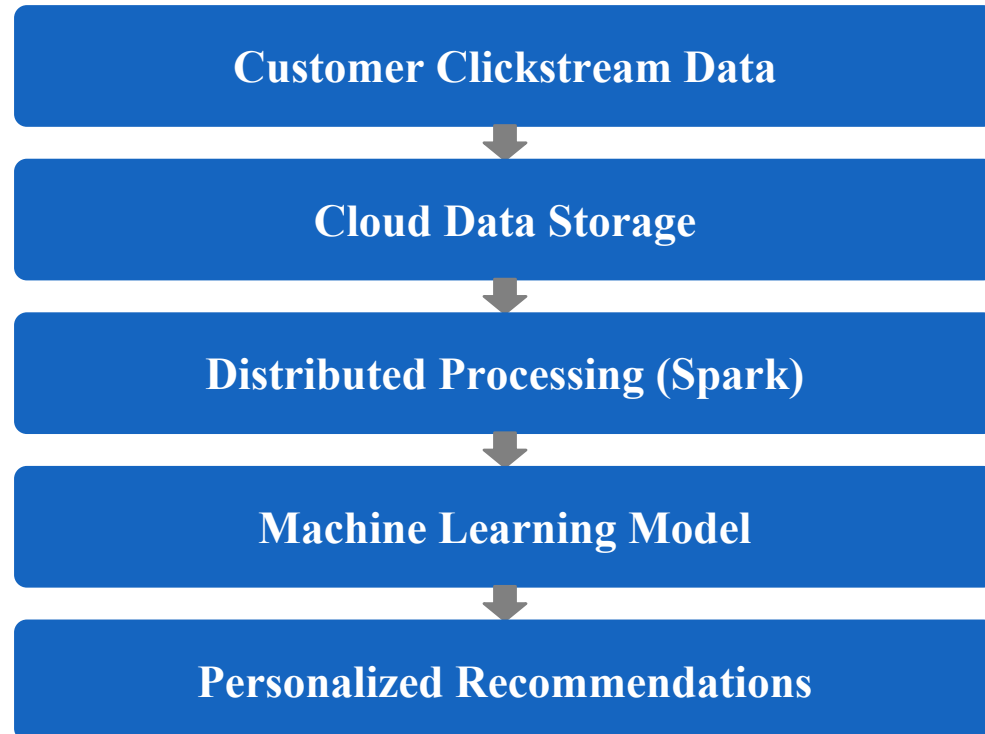
Predictive maintenance
Supply chain optimization

Social Media

Sentiment analysis
Trend detection

Example Use Case

E-Commerce Recommendation System



Advantages

- Higher customer engagement.
- Increased sales conversion.
- Improved customer experience.
- Data-driven business decisions.



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Conclusion and Key Takeaways

Cloud computing gives big data analytics the storage, processing power, and scalability needed to turn raw data into actionable insight.

Key Takeaways

- Big data is defined by volume, velocity, variety, veracity, and value.
- Cloud platforms provide elastic storage and distributed processing.
- Integration enables scalability, cost efficiency, and real-time analytics.
- Applications span retail, healthcare, finance, and manufacturing.
- Data security and privacy remain ongoing challenges.



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Limitations and Future Scope

Limitations

- Data privacy and security risks in shared cloud environments.
- High cost of large-scale data transfer and storage over time.
- Dependency on stable internet connectivity.
- Shortage of skilled big data and cloud professionals.

Future Scope

- Tighter integration of AI and edge computing with cloud analytics.
- Growth of real-time streaming analytics platforms.
- Stronger automated data governance and security frameworks.



Engineer to Excel



Thank You!!!